DEPARTMENT OF SOFTWARE ENGINEERING
BACHELORS IN SOFTWARE ENGINEERING

Course Code: CT-317 | Course Title: Machine Learning
SE Batch 2023, Fall Semester 2026

Complex Engineering Problem

Sentiment Analysis from Audio Recordings

Speech emotion and sentiment recognition with noise filtering, self-supervised speech features and few-shot personalisation

Group Members

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Source code: `https://github.com/hbnasir23/ml-cep-ct317`

Submitted to: _____

Date of submission: _____

Abstract

This project builds a system that recognises the emotion and sentiment of a speaker from a short voice recording. We use 7,442 noisy clips of 91 actors from CREMA-D (six emotions: anger, disgust, fear, happy, neutral, sad). The clips are de-noised with a 4 kHz low-pass filter and evaluated on a strict *speaker-independent* split: the test speakers are never heard during training.

We compare two families of features with the five classifiers required by the handout (KNN, Logistic Regression, Naive Bayes, SVM and a Neural Network). The first family is 242 hand-crafted features from the literature (FFT bands, MFCCs, mel-spectrogram, chromagram, spectral shape, prosody, bicoherence); the best classifier with them, an SVM, reaches **52.4%** six-emotion accuracy. The second family is embeddings from the pretrained self-supervised speech model WavLM; these raise accuracy to **68.9%** (Logistic Regression), and three-class sentiment (positive, neutral, negative) is recognised with **86.4%** accuracy.

Testing on a second corpus (RAVDESS) exposed the main practical problem, a loss of accuracy on new recording conditions, which is also why our first model performed poorly on our own microphone recordings. Two remedies were developed and measured:

- *multi-corpus training*, which lifts accuracy on unseen RAVDESS actors from 51% to **70.5%** without losing CREMA-D accuracy;
- *few-shot personalisation*, in which a new user provides only two labelled clips per emotion; this raises accuracy on their other recordings from 71% to **82%** (emotion) and from 84% to **90%** (sentiment).

All of this is available in an interactive web application with live microphone input, input-quality warnings and in-app personalisation. The code is public at `https://github.com/hbnasir23/ml-cep-ct317`.

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Chapter 1

Introduction

1.1 Problem statement

Sentiment analysis from audio uses the acoustic properties of speech, such as intonation, pitch, energy and tempo, to infer the emotional state of a speaker. It has applications in call-centre quality monitoring, voice assistants, media analysis and mental-health support. The CEP asks us to:

1. select audio files and create training, validation and test sets that are best suited to train and evaluate the classifiers;
2. remove noise from the audio using low-pass filtering;
3. review the literature and extract suitable features (Fourier transform, MFCC, mel-spectrogram, chromagram, bicoherence, spectral centroid, bandwidth and contrast);
4. evaluate K-Nearest Neighbours, Logistic Regression, Naive Bayes, Support Vector Machines and a Neural Network (scikit-learn).

1.2 Motivation for the extended work

Our first complete pipeline met all four requirements but reached only about 52% accuracy on unseen speakers. It also behaved poorly when we tested it with *our own voices* through a microphone. We therefore set ourselves a further goal: build a model that remains accurate on recordings it was not trained on, and measure this honestly. This led to three additions: (i) pretrained self-supervised speech features, (ii) cross-corpus and multi-corpus evaluation, and (iii) few-shot personalisation to the user's own voice.

1.3 Contributions

- A complete, reproducible pipeline covering data selection, low-pass de-noising, feature extraction and five classifiers on a speaker-independent split.
- A hand-crafted feature baseline (242 features) with a feature-group ablation and a study of the filter cut-off.
- WavLM layer embeddings that raise emotion accuracy from 52.4% to 68.9% and sentiment accuracy to 86.4%.
- Cross-corpus testing on RAVDESS, and multi-corpus training that improves generalisation to new recording conditions.
- Few-shot personalisation, measured on eight unseen speakers.

- An interactive web application: microphone input, sentiment and emotion with probabilities, a view of what the filter removed, input-quality checks, and personalisation in the browser.

1.4 Tools

Python 3.13, scikit-learn 1.8 [1], librosa 1.0 [2], SciPy, PyTorch 2.10 and Hugging Face Transformers (WavLM), and Streamlit for the application. Everything runs on a CPU.

Chapter 2

Datasets and Data Selection

2.1 CREMA-D (main dataset)

The provided repository contains a noisy version of CREMA-D [3]: **7,442** WAV clips (16 kHz, mono, about 2–3 s) from **91 actors** (48 male, 43 female, aged 20–74, of diverse ethnicities). Each actor reads 12 sentences with one of six emotions. File names encode the metadata, for example `1001_DFA_ANG_XX.wav` = actor 1001, sentence "DFA", *Anger*, unspecified intensity. The classes are almost balanced: 1,271 clips each for anger, disgust, fear, happy and sad, and 1,087 for neutral.

2.2 Speaker-independent split

A classifier that will be used on *new* people must be evaluated on people it has never heard. If clips were split at random, every actor would appear in both training and test data, and part of the accuracy would come from recognising the voice instead of the emotion. We therefore split the **actors** 70/15/15 with `GroupShuffleSplit`, and the code asserts that the three actor sets are disjoint (Table 2.1). All emotions and all 12 sentences appear in every subset. Validation speakers are used for all model choices; the test speakers are used exactly once per reported result.

Table 2.1: Speaker-independent split of CREMA-D.

Split	Actors	ANG	DIS	FEA	HAP	NEU	SAD	Total
Train	63	880	880	880	880	752	880	5,152
Validation	14	195	195	195	195	167	195	1,142
Test	14	196	196	196	196	168	196	1,148

2.3 RAVDESS (second corpus)

To test robustness to new recording conditions we use RAVDESS [4]: 24 professional actors (12 female, 12 male), recorded in a different studio with different equipment, as clean 48 kHz audio (resampled to 16 kHz). It covers the same six emotions plus *calm* and *surprised*, which we exclude, leaving **1,056 clips**. It is used in two ways: (a) as a completely unseen test corpus, and (b) split by actor (actors 1–16 for training, 17–24 for testing) for the multi-corpus experiments.

2.4 Sentiment labels

The handout speaks of *sentiment*. Following the usual valence grouping, we map the six emotions to three sentiments: **Positive** = Happy, **Neutral** = Neutral, and **Negative** = Anger, Disgust, Fear, Sad. The sentiment probabilities are the sums of the corresponding emotion probabilities. Because four of the six emotions are negative, always answering "Negative" already gives 68.3% sentiment accuracy on the test speakers. We therefore also report *balanced accuracy* and macro-F1, for which that trivial baseline scores 33.3%.

Chapter 3

Pre-processing: Noise Removal

Every clip, including microphone recordings in the application, goes through the same three steps:

1. **Low-pass filtering.** A 6th-order Butterworth low-pass filter with a **4 kHz** cut-off is applied in second-order sections with forward–backward filtering (`sosfiltfilt`). Forward–backward filtering has zero phase distortion, so the timing of pitch and energy contours is preserved. Most of the prosodic and emotional information in speech lies below 4 kHz [5], whereas the added noise extends up to the 8 kHz Nyquist limit.
2. **Silence trimming** of leading and trailing segments more than 30 dB below the peak.
3. **Peak normalisation**, so that the recording gain and the distance to the microphone do not matter.

The filter also makes clean microphone recordings and the noisy training clips more alike, since both are reduced to the same 0–4 kHz band.

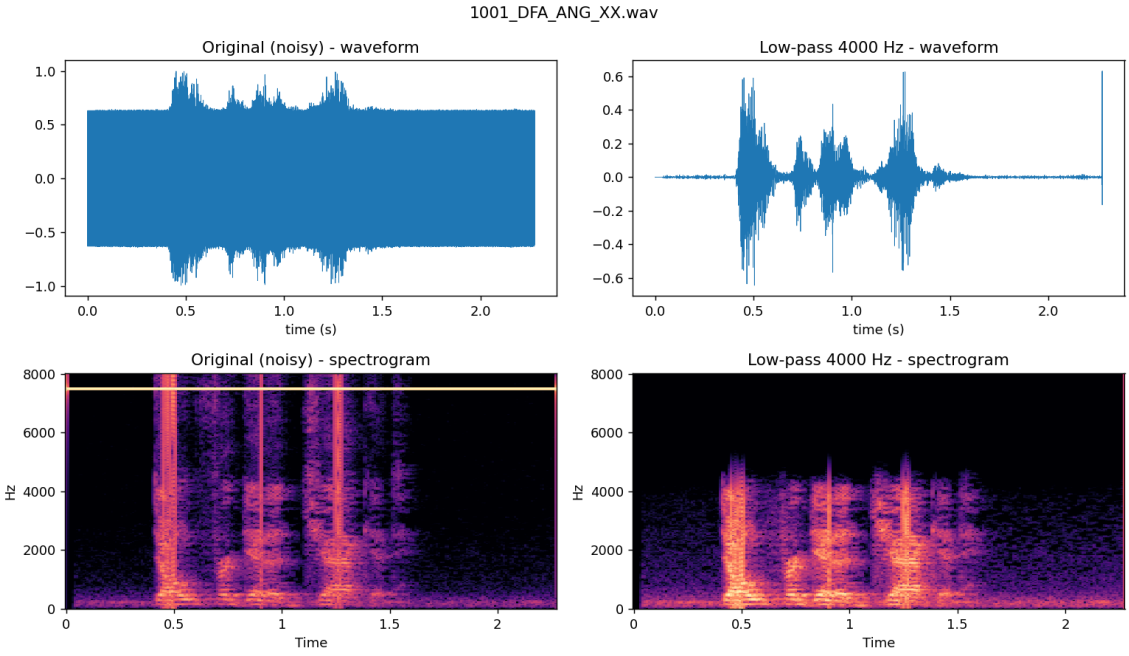


Figure 3.1: Waveform (top) and spectrogram (bottom) of a noisy clip before and after the 4 kHz low-pass filter.

Chapter 4

Feature Extraction

4.1 Literature review

Speech emotion recognition has traditionally relied on hand-crafted descriptors [6, 7]:

- *Spectral and cepstral* features, which describe the vocal-tract shape and voice quality: MFCCs, mel-spectrogram, spectral centroid, bandwidth, contrast, roll-off and flatness.
- *Prosodic* features (fundamental frequency F0, energy, speaking rate), which mainly carry arousal. Anger and happiness, for example, have higher and more variable pitch and energy than sadness.
- *Higher-order spectra* such as the bicoherence, which measures quadratic phase coupling and has been used for synthetic-speech detection [8].

Since 2020, *self-supervised* speech models such as wav2vec 2.0 [9], HuBERT and WavLM [10] have been pretrained on tens of thousands of hours of unlabelled speech. Their intermediate layers encode speaker, phonetic and paralinguistic information, and on the SUPERB benchmark [11] a light classifier on frozen WavLM features outperforms hand-crafted feature sets for emotion recognition. We evaluate both approaches.

4.2 Hand-crafted features (baseline)

Frame-level features use a 32 ms window and a 10 ms hop. They are summarised over time by mean and standard deviation, giving **242** values per clip (Table 4.1).

Table 4.1: Hand-crafted feature groups.

Group	Dim.	Description
Fourier transform	16	log relative energy of the FFT magnitude in 16 bands (0–4 kHz)
MFCC	80	20 MFCCs and their deltas (mean, std)
Mel-spectrogram	80	40 log-mel bands (mean, std)
Chromagram	24	12 pitch classes (mean, std)
Spectral shape	22	centroid, bandwidth, roll-off, flatness, 7-band contrast
Prosody	12	F0 (YIN) statistics, RMS energy, zero-crossing rate, voiced ratio, duration
Bicoherence	8	mean, variance, skewness, kurtosis of magnitude and phase

The bicoherence is estimated by averaging over Hann-windowed segments:

$$b(f_1, f_2) = \frac{|\mathbb{E}[X(f_1)X(f_2)X^*(f_1+f_2)]|}{\sqrt{\mathbb{E}[|X(f_1)X(f_2)|^2]\;\mathbb{E}[|X(f_1+f_2)|^2]}}. \tag{4.1}$$

4.3 Self-supervised features: WavLM

WavLM-base-plus [10] (94 M parameters, pretrained by Microsoft on 94k hours of unlabelled speech) consists of a convolutional encoder followed by 12 transformer layers. We feed it the filtered 16 kHz signal and average the output of each of the 13 layers over time, giving 13 vectors of 768 numbers per clip. The network is *frozen*: it is used only as a feature extractor, and the actual classifiers are the scikit-learn models required by the handout. Extracting features for all clips took about 45 minutes on a 12-core CPU.

Choosing the layer. Different layers encode different information, so a quick linear classifier (Logistic Regression) was trained on each layer and scored on the *validation* speakers (Figure 4.1). Emotion information peaks in the middle of the network (**layer 7**, 71.3% validation accuracy). Early layers are closer to the raw spectrum, and the last layers specialise in the pretraining objective. Layer 7 is used for all further experiments.

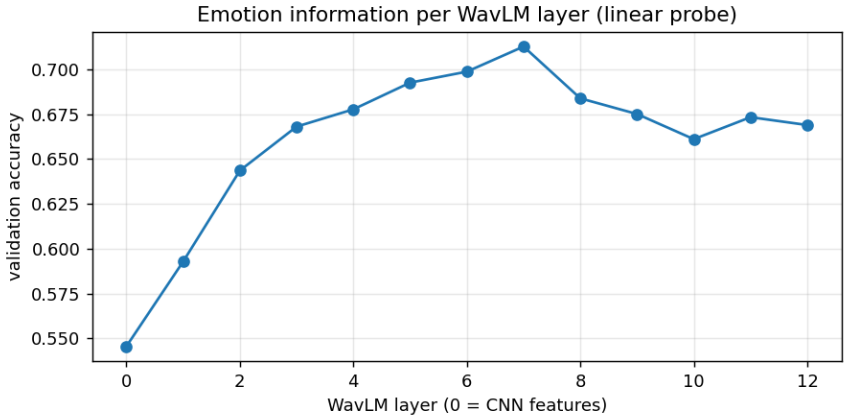


Figure 4.1: Validation accuracy of a linear classifier on each WavLM layer.

Chapter 5

Classifiers and Training Protocol

Each classifier follows a `StandardScaler` fitted on the training data only. For Naive Bayes on WavLM features, a PCA whitening step (128 components) is added, because Gaussian NB assumes independent features. Hyper-parameters are selected by grid search on the validation speakers; the best setting is refit on training and validation data, and the test speakers are evaluated once.

Table 5.1: Hyper-parameter grids (WavLM features; the baseline grids are similar).

Classifier	Grid
KNN	$k \in \{5, 11, 21, 41\}$, uniform or distance weights, Euclidean or cosine
Logistic Regression	multinomial, L2, $C \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$
Gaussian Naive Bayes	variance smoothing $\in \{10^{-9}, 10^{-6}, 10^{-3}, 10^{-1}\}$
SVM (RBF)	$C \in \{1, 3, 10, 30\}$, $\gamma \in \{\text{scale}, 3 \cdot 10^{-4}, 10^{-3}\}$
Neural Network (MLP)	hidden layers (256), (512), (512,128); $\alpha \in \{0.01, 0.1, 1\}$; Adam, early stopping

Chapter 6

Results

6.1 Baseline: hand-crafted features

Table 6.1: Six-emotion accuracy on unseen CREMA-D speakers with hand-crafted features (chance = 16.7%).

Model	Selected hyper-parameters	Val. acc.	Test acc.	Test macro-F1
KNN	$k = 15$, cosine	0.469	0.408	0.381
Logistic Regression	$C = 1$	0.525	0.519	0.519
Naive Bayes	smoothing 10^{-9}	0.387	0.402	0.354
SVM (RBF)	$C = 3$, γ=scale	0.531	**0.524**	**0.522**
Neural Network	(128), $\alpha = 0.1$	0.537	0.513	0.513

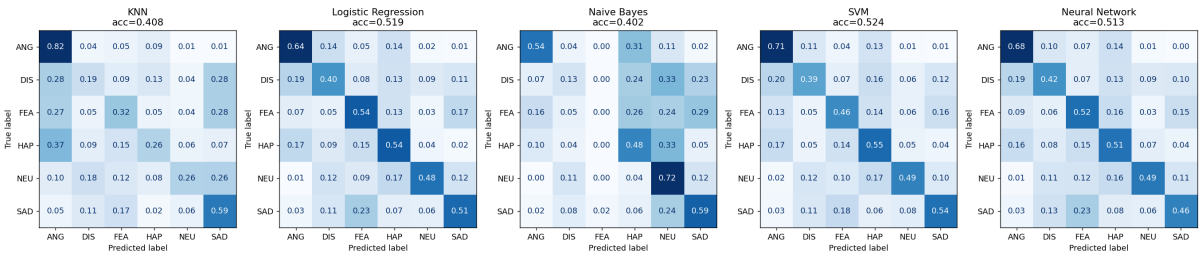


Figure 6.1: Hand-crafted features: row-normalised test confusion matrices.

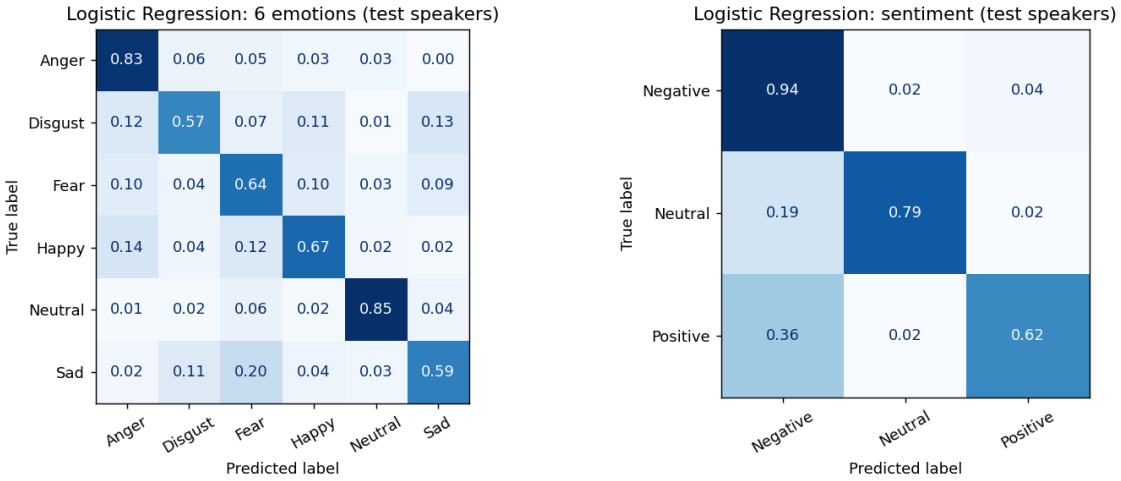
Feature ablation and filter cut-off. With the SVM, MFCCs are the strongest single group (0.490 on their own), and removing them causes the largest drop (to 0.504). Prosody is remarkably strong for only 12 numbers (0.449). Bicoherence (0.289) and chroma (0.351) are weak on their own and redundant in the full set. Varying the low-pass cut-off between 2 kHz and no filter changed accuracy only within 0.524–0.531, which confirms that the emotional cues lie below 3–4 kHz.

6.2 WavLM features

Table 6.2: WavLM layer-7 features on unseen CREMA-D speakers. Sentiment majority baseline: accuracy 0.683, balanced accuracy 0.333.

	6 emotions			Sentiment (3 classes)		
Model	Val.	Test	F1	Test	F1	Bal. acc.
KNN ($k = 5$, cosine, distance)	0.535	0.528	0.513	0.789	0.681	0.667
Logistic Regression ($C = 0.01$)	**0.713**	**0.689**	**0.692**	**0.864**	**0.810**	**0.785**
Naive Bayes (PCA-128)	0.519	0.531	0.527	0.742	0.663	0.696
SVM ($C = 10$, $\gamma = 3 \cdot 10^{-4}$)	0.711	0.681	0.683	0.861	0.800	0.769
Neural Network (512, $\alpha = 1$)	0.703	0.677	0.679	0.860	0.801	0.770

Self-supervised features improve every classifier, and the best result rises from 52.4% to **68.9%** (+16.5 points). Logistic Regression, the simplest discriminative model, is the best. This is typical for strong pretrained embeddings, which make the classes close to linearly separable. KNN and Naive Bayes remain weak: KNN suffers from the 768-dimensional space and from distances dominated by speaker differences, and Naive Bayes from its independence assumption.



(a) Six emotions

(b) Sentiment

Figure 6.2: Logistic Regression on WavLM features: row-normalised confusion matrices on the unseen CREMA-D test speakers.

Per-class behaviour. Neutral (F1 0.86) and Anger (F1 0.75, recall 0.83) are the easiest classes. Fear (0.60) and Disgust (0.62) are the hardest and are mostly confused with Sad and Anger respectively. For sentiment, Negative is recognised with recall 0.94 and Neutral with 0.79. Positive (recall 0.62) is sometimes mistaken for Negative, because Happy and Anger share high arousal (loud, high-pitched speech).

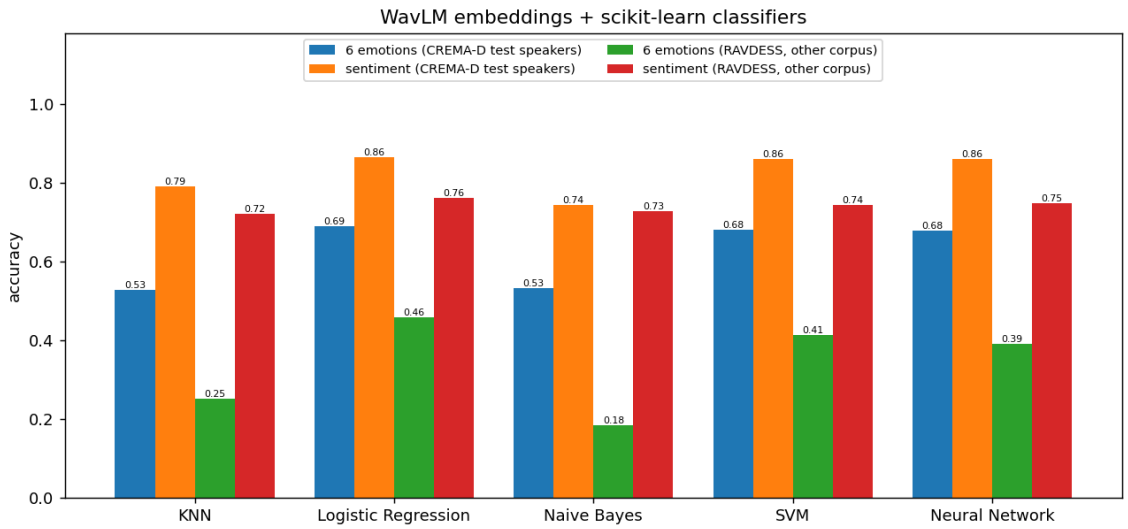


Figure 6.3: All five classifiers on WavLM features: CREMA-D test speakers and the unseen RAVDESS corpus.

6.3 Generalisation to a new corpus

RAVDESS was never used for training in the experiments of Table 6.2. On it the best model reaches only **45.7%** emotion accuracy and 57.1% balanced sentiment accuracy (Figure 6.3). A different studio, microphone and acting style is enough to cut performance by a third. This *domain shift* is exactly what we experienced when testing with our own voices, and it is invisible if one evaluates only on a split of a single dataset.

6.4 Remedy 1: multi-corpus training

We added RAVDESS actors 1–16 to the training data and tested on the eight RAVDESS actors 17–24 that were never seen (Table 6.3).

Table 6.3: Effect of multi-corpus training (Logistic Regression, WavLM layer 7).

	CREMA-D test speakers		RAVDESS actors 17–24	
Training data	Emotion	Sentiment	Emotion	Sentiment
CREMA-D only	0.689	0.864	0.514	0.787
CREMA-D + RAVDESS 1–16	**0.693**	0.860	**0.705**	**0.852**

Accuracy on new RAVDESS speakers rises by 19 points, while CREMA-D accuracy is unchanged. More diverse training data makes the model more robust, so the deployed model is trained on both corpora (all 8,498 clips).

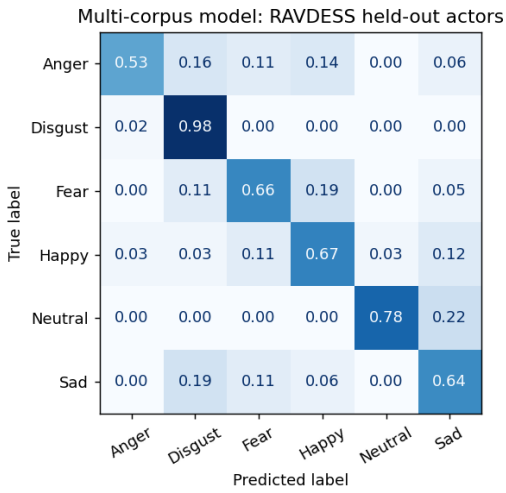


Figure 6.4: Multi-corpus model on the unseen RAVDESS actors 17–24.

6.5 Remedy 2: few-shot personalisation

Even a multi-corpus model sees a new user through a new microphone, room and speaking style. Our solution is to let the user provide a few labelled examples of their own voice. The base model is then retrained on the corpus plus these examples, each user example carrying a sample weight of 20. With Logistic Regression this takes a few seconds, so it can be done live in the application.

Evaluation. We simulated eight new users with RAVDESS actors 17–24, whom the base model never saw. For each actor, k randomly chosen clips per emotion were used as the user's examples, and the actor's *remaining* clips formed the test set. Each experiment was repeated with three random draws (Table 6.4, Figure 6.5).

Table 6.4: Few-shot personalisation: mean accuracy (± std over actors and draws) on each speaker's other clips.

Own clips provided	Emotion accuracy	Sentiment accuracy
0 (base model)	0.710 ± 0.072	0.844 ± 0.060
6 ($k = 1$ per emotion)	0.782 ± 0.088	0.875 ± 0.061
12 ($k = 2$)	**0.824** ± 0.092	**0.896** ± 0.066
18 ($k = 3$)	0.829 ± 0.092	0.885 ± 0.070

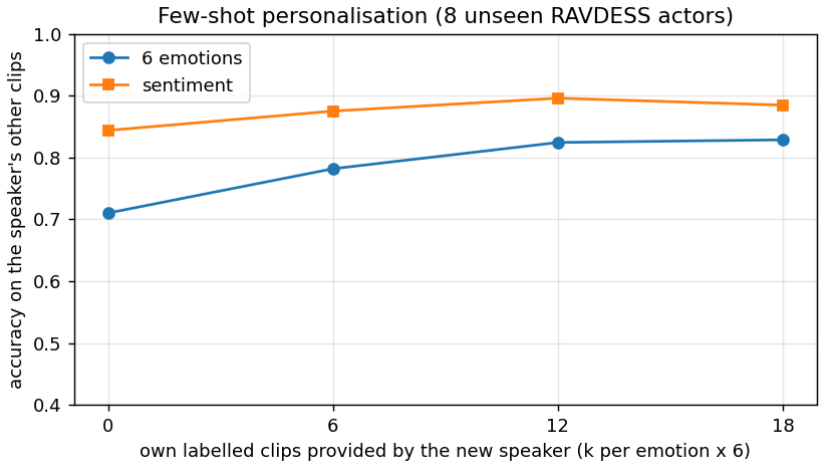


Figure 6.5: Accuracy as a function of the number of the user's own labelled clips.

Twelve short recordings (two per emotion) raise emotion accuracy by **11 points to 82%** and sentiment accuracy to **90%**. Beyond that, the gain levels off.

6.6 Summary of results

Table 6.5: Progression of the system (emotion accuracy on speakers not seen in training).

System	Same corpus (CREMA-D)	New corpus (RAVDESS)
Hand-crafted features + SVM	0.524	–
WavLM + Logistic Regression	0.689	0.457
+ multi-corpus training	0.693	0.705
+ 12 own clips (personalised)	–	**0.824**

Chapter 7

Innovation: Interactive Application with Personalisation

The final model is delivered as a web application (`streamlit run app.py`). It covers the whole path from a person speaking into a microphone to an explained prediction.

7.1 Features

- **Live microphone input.** The user records a sentence in the browser; uploads, CREMA-D test clips and RAVDESS clips are also available, the latter two with their true label shown for comparison.
- **Sentiment and emotion.** The prediction is shown as a colour-coded sentiment (positive, neutral, negative) and an emotion, each with its probability distribution and a low-confidence notice when no emotion clearly dominates.
- **What the filter removed.** The spectrogram of the input is shown before and after the 4 kHz low-pass filter.
- **Input-quality checks.** Recordings that are too short, too long, too quiet or clipped trigger a warning. These are the most common reasons for wrong predictions on home recordings.
- **Feedback and few-shot personalisation.** After each prediction the user can state the emotion they actually expressed and add the clip as an example. A progress table shows how many examples exist per emotion, and *Personalise* retrains the classifier on the corpus plus the user's examples in a few seconds (Section 6.5). The user can switch between the base and personalised models and compare them in a session history.

7.2 Why this is innovative

Most student emotion-recognition projects stop at an accuracy number on a random split of a single dataset. This project (i) evaluates on unseen speakers *and* an unseen corpus, (ii) identifies domain shift as the reason a model fails on real users, and (iii) gives the user a practical, measured way to fix it, few-shot personalisation, with immediate visual feedback. The application turns the model into a tool that a non-expert can test, understand and adapt to their own voice.

Chapter 8

Software Design and Usage

8.1 Structure

```
ml_cep/
  app.py                      interactive application
  q2/audio_features.py        low-pass filter + 242 hand-crafted features
  q2/q2_audio_sentiment.py    split, baseline classifiers, ablation, cut-off study
  q2/ssl_features.py          WavLM embedding extraction
  q2/q2_wavlm_emotion.py      layer choice, 5 classifiers, RAVDESS, multi-corpus, final
     model
  q2/personalization.py       few-shot personalisation (experiment + function used by
    app)
  q2/predict.py               command-line prediction for WAV files
  q2/models/                  trained models (+ corpus features for personalisation)
  q2/results/                 all tables and figures
  report/                     this report
```

8.2 How to run

```
pip install -r requirements.txt
git clone https://github.com/naweedfatima03/SentimentAnalysisDataset q2/dataset_repo
# RAVDESS speech: zenodo.org/records/1188976 -> extract to q2/ravdess/
python q2/q2_audio_sentiment.py        # split + baseline
python q2/ssl_features.py ...          # (called per shard, see README)
python q2/q2_wavlm_emotion.py          # main model
python q2/personalization.py           # personalisation experiment
python q2/predict.py my_voice.wav      # predict a file
streamlit run app.py                   # application
```

The trained models are included in the repository, so `predict.py` and the application work right after cloning. WavLM itself (about 380 MB) is downloaded automatically on first use. Random seeds are fixed (42), and features are cached so that every result in this report can be regenerated.

Chapter 9

Team Contribution

Table 9.1: Individual contributions.

Member	**Main responsibilities**
Sumayyah Mohammedy (SE-23013)	Literature review of speech emotion recognition and self-supervised speech models; dataset analysis, speaker-independent split and sentiment mapping; compilation and editing of the LaTeX report.
Amna Khan (SE-23039)	Pre-processing (low-pass filter design, silence trimming, normalisation) and the cut-off study; the 242 hand-crafted features including bicoherence; baseline classifiers and feature-group ablation.
Mariha Mahmood (SE-23047)	WavLM embedding extraction, layer-wise analysis, tuning and evaluation of the five classifiers on WavLM features, sentiment evaluation with balanced metrics, confusion and error analysis.
Haris Bin Nasir (SE-23084)	RAVDESS cross-corpus test, multi-corpus training, few-shot personalisation (method and experiment), the Streamlit application, command-line tools and GitHub repository.

Table 9.2: Responsibility matrix (**L** = lead, **S** = support).

Task	Sumayyah	Amna	Mariha	Haris
Literature review, data and split	L	S		
Filtering and hand-crafted features		L	S	
WavLM features and classifiers		S	L	
Cross-corpus and personalisation			S	L
Application and repository	S			L
Report	L	S	S	S

Chapter 10

Conclusion

We built a speech emotion and sentiment recogniser that meets all requirements of the CEP: data selection with a speaker-independent split, low-pass de-noising, literature-based features, and the five required classifiers. Hand-crafted features give 52.4% six-emotion accuracy on unseen speakers. Frozen WavLM features with Logistic Regression give **68.9%**, and sentiment is recognised with **86.4%** accuracy.

Cross-corpus testing showed that the main obstacle in practice is domain shift. Multi-corpus training (70.5% on unseen RAVDESS actors, up from 51.4%) and few-shot personalisation (82% emotion and 90% sentiment accuracy with 12 of the user's own clips) address it. The interactive application makes both the predictions and the personalisation available to any user.

Limitations. Both corpora contain acted English speech, and spontaneous real-life emotions are subtler. Six-way emotion classification remains ambiguous even for human listeners (CREMA-D raters agree with the intended emotion in only about 64% of audio-only cases [3]).

Future work. Fine-tuning WavLM end to end, larger models (WavLM-large), more corpora (IEMOCAP, MSP-Podcast) and data augmentation with real room and microphone responses.

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Appendix A

Requirement Checklist

Table A.1: CEP requirements and where they are addressed.

Requirement	Where
Select audio files; create training, validation and test sets	Ch. 2, Table 2.1
Remove noise using low-pass filtering	Ch. 3, Fig. 3.1
Literature review and features (FFT, MFCC, mel, chroma, bicoherence, centroid, bandwidth, contrast)	Ch. 4, Table 4.1
KNN, Logistic Regression, Naive Bayes, SVM, Neural Network	Ch. 5–6, Tables 6.1, 6.2
Project report, code, ease of use	this report, `https://github.com/hbnasir23/ml-cep-ct317`, Ch. 7